

# 1 Rollout

policy

$\pi_t$



transitions

GAE

$A_r, A_{c_k}$

episode cost

$\bar{J}_k$

# 2 Flow inputs

state clusters



transition edges



$\hat{F}$   
flow



advantage



$C_k$   
costs

# 3 Set constraints

Soft constraints

all cost budgets



Hard constraints

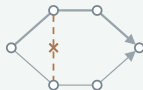


remove edges

# 4 Entropic projection

One flow solve

$\hat{F}$



$F^*$



target flow  
is not executed

prices go to the actor

# 5 Priced policy update

Safety prices

clip + EMA

Observed-cost feedback

cost - budget

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO/TRPO  $\rightarrow \pi_{t+1}$

observed episode costs

GAE advantages

next rollout